THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET B

Chapters 5 & 7: **Arithmetic Progressions, Coordinate Geometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The next term of the AP $5, 2, -1, \ldots$ is: 1

(a) 4 **(b)** −2 **(c)** −3 **(d)** −4

2. If $a_n = 2n + 1$, the first three terms are: 1

(a) $1, 3, 5$ **(b)** $2, 4, 6$ **(c)** $3, 4, 5$ **(d)** $3, 5, 7$

3. The number of terms in the AP $10, 13, \ldots, 100$ is: 1

(a) 30 **(b)** 31 **(c)** 29 **(d)** 32

4. A student states: "If a, b, c are in AP, then $a + c = b$." The correct response is: 1

(a) Correct **(b)** $a + b = c$

(c) Incorrect — $a + c = 2b$ **(d)** Cannot be determined

5. For the sum of the first 15 terms of $2, 4, 6, \ldots$, Student A gets 240 and Student B gets 225. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

6. The distance between $(2, -3)$ and $(-2, 0)$ is: 1

(a) 25 **(b)** 5 **(c)** $\sqrt{7}$ **(d)** 7

7. The point $(4, -6)$ lies in the: 1

(a) second quadrant **(b)** fourth quadrant **(c)** first quadrant **(d)** third quadrant

8. If the mid-point of the join of $(a, 4)$ and $(2, 2b)$ is $(3, 5)$, then (a, b) is: 1

(a) $(4, 3)$ **(b)** $(4, 6)$ **(c)** $(3, 4)$ **(d)** $(2, 3)$

9. A student states: "The distance between (x_1, y_1) and (x_2, y_2) is $(x_2 - x_1) + (y_2 - y_1)$." The correct response is: 1

(a) Correct for positive coordinates **(b)** Correct

(c) Incorrect — it is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ **(d)** Cannot be determined

10. The y-axis divides the join of $(-2, 3)$ and $(4, 1)$ in a ratio: Student A says 1 : 2; Student B says 2 : 1. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Which term of the AP $5, 15, 25, \ldots$ will be 130 more than its 31st term? **2**

12. Find the sum of the first 40 positive integers divisible by 6. **2**

13. Find the distance of the point $(-5, 12)$ from **2**
(i) the origin
(ii) the point $(0, 12)$.

14. The mid-point of the segment joining A(3, 4) and B(k, 6) is (4, 5). Find k. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the sum: $-5 + (-8) + (-11) + \cdots + (-230)$. **3**

16. Find the ratio in which the x-axis divides the segment joining A(6, 3) and B(−2, −5), and the point of division. **3**

17. If the 9th term of an AP is zero, prove that its 29th term is twice its 19th term. **3**

OR

Find the 12th term from the end of the AP $3, 5, 7, \ldots, 201$. **3**

18. Find the type of triangle formed by A(−2, 0), B(2, 0) and C(0, 2). Justify. **3**

OR

Find the point on the y-axis equidistant from A(−4, 3) and B(5, 2). **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A man repays a loan of ₹3250 by paying ₹20 in the first month and then increasing the payment by ₹15 every month. **5**
(i) Form the equation for the number of months n.
(ii) Find n.
(iii) Find the last instalment.

OR

The first and last terms of an AP are 17 and 350, and the common difference is 9. **5**
(i) How many terms are there and what is their sum?
(ii) Find the two middle terms.

20. **Case study:** On a city map, a home is at H(1, 2) and a school at S(7, 10). **5**
(i) Find HS.
(ii) A shop lies on HS dividing it in the ratio 2 : 3 from H. Find the shop's position.
(iii) A friend's house F is such that S is the mid-point of HF. Find F.
(iv) Find HF.

OR

Show that $(7, 10)$, $(-2, 5)$ and $(3, -4)$ are the vertices of an isosceles right triangle. Show that the mid-point of the hypotenuse is equidistant from all three vertices. **5**

END OF QUESTION PAPER — **Total: 40 Marks**